MILITARY
SPECIFICATION

MIL-T-2534B(SigC)
22 August 1958

Superseding MIL-T-2534A(SigC)
26 March 1952
and Amendment No. 1,
4 March 1957

# TRAINING GENERATOR AN/URA-TI( )
## (Units of)

## 1. SCOPE

1.1 Scope.- This specification covers technical requirements, and means for determining compliance therewith, of the following units of Training Generator AN/URA-TI( ):

(a) Modulator MD-22( )/URA-TI
(b) Cable Assembly, special purpose, electrical, CX-2950/U (10 ft. 0 in.)
(c) Cable Assembly, special purpose, electrical, CX-2951/U (10 ft. 0 in.)
(d) Cable Assembly, power, electrical, CX-3099/U (10 ft. 0 in.)
(e) Cable Assembly, power, electical, CX-111/URA-TI
(f) Mounting FT-250-( )
(g) Cover BC-185( )

This specification also covers requirements for preparation for delivery of the above items and ancillary items and running spare parts of the Training Generator, and means for determining compliance with such requirements.

## 2. APPLICABLE DOCUMENTS

2.1 The following specification, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| T-T-871 | Twine, cotton, wrapping |
| TT-C-595 | Colors; (for) ready-mixed paints |
| UU-T-111 | Tape; paper, gummed (sealing and securing) |
| LLL-F-291 | Fiberboard; corrugated, single face(flexible) |
| PPP-B-621 | Boxes, wood, nailed and lock-cover. |
| PPP-B-636 | Boxes, fiber |
| PPP-T-76 | Tape, pressure-sensitive adhesive, paper, water resistant. |

MILITARY

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|---|---|
| MIL-E-1 | Electron tubes |
| MIL-P-75 | Packaging, packing, and container marking of electron tubes |



FSC-6940

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|---|---|
| MIL-P-116 | Preservation, methods of |
| MIL-B-117 | Bags, interior, packaging |
| MIL-T-152 | Treatment, moisture-and-fungus-resistant, of communications, electronic and associated electrical equipment. |
| MIL-C-3432 | Cable, flexible and extra flexible, 300 and 600 volts |
| MIL-C-3767 | Connectors (electrical, power bladed type) |
| MIL-C-3883 | Cord, electrical (audio frequency) |
| MIL-C-3884 | Cord, electrical (short lay) |
| MIL-C-3885 | Cable assemblies and cord assemblies, electrical (for use in electronic, communications, and associated electrical equipment |
| MIL-P-11268 | Parts, materials, and processes used in Signal Corps equipment. |
| MIL-I-11748 | Interference reduction for electrical and electronic equipment. |
| MIL-M-13231 | Marking of electronic items |
| MIL-F-14072 | Finishes for ground signal equipment. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling procedures and tables for inspection by attributes |
| MIL-STD-129 | Marking for shipment and storage |
| MIL-STD-169 | Extreme temperature cycle |
| MIL-STD-170 | Moisture resistance test cycle for ground signal equipment. |
| MIL-STD-200 | Electron tubes |
| MIL-STD-252 | Wired equipment, classification of visual and mechanical defects |

DRAWINGS

SIGNAL CORPS

| | |
|---|---|
| SC-A-21825 | Identification tags |
| SC-C-33073 | Pilot pack process sheet |
| SC-DL-184115 | Modulator MD-22( )/URA-TI |
| SC-DL-184371 | Cable assembly, special purpose, electrical, CX-2950/U |
| SC-DL-184372 | Cable assembly, special purpose, electrical, CX-2951/U |
| SC-DL-184373 | Cable assembly, power, electrical, CX-3099/U |
| SC-DL-184374 | Cable assembly, power, electrical, CX-111/URA-TI |
| SC-DL-14643 | Mounting FT-250-( ) |
| SC-D-12631 | Cover BG-185( ) |

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FOR RAIL TRANSPORTATION:

Uniform Freight Classification No. 4

(Application for copies should be addressed to Uniform Freight Classification, 202 Union Station Chicago 6, Illinois.)

Consolidated Freight Classification Rules No. 21

(Application for copies should be addressed to the Consolidated Classification Committee, 1 Park Avenue and 33rd Street, New York 16, New York.)

FOR TRUCK TRANSPORTATION:

National Motor Freight Classification No. A-3 and No. 14

(Application for copies should be addressed to American Trucking Association, Inc., 1424 Sixteenth Street, Washington 6, D. C.)

3. REQUIREMENTS

3.1 Description.- The Training Generator,AN/URA-TI( ), is a portable device, which produces variable tone (bagpipes), noise, or random CW and MCW signals, which may be used to modulate or key any communications transmitter to produce intentional interference signals.

3.2 Construction.- The equipment shall be constructed in accordance with the following drawings and drawing and data lists:

(a) SC-DL-184115 for Modulator MD-22( )/URA-TI
(b) SC-DL-184371 for Cable Assembly CX-2950/U
(c) SC-DL-184372 for Cable Assembly CX-2951/U
(d) SC-DL-184373 for Cable Assembly CX-3099/U
(e) SC-DL-184374 for Cable Assembly CX-111/URA-TI
(f) SC-DL-14643 for Mounting MT-250-( )
(g) SC-D-12631 for Cover BG-185( )

3.3 Parts, materials, and processes; general.- In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.



(2) The sole

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3.4 Tropicalization.-

3.4.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.4.2 unless the material has been given the treatment specified in Specification MIL-P-11268 or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.4.2 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.4.3 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This statement shall include the statements required by Specifications MIL-P-11268 and MIL-T-152, as applicable.

3.5 Tubes, electron.- Electron tubes shall conform to Specification MIL-E-1 and shall be of the types given in Standard MIL-STD-200. Each equipment that uses electron tubes shall be furnished with the same tubes used when the equipment was inspected. Unless separate packaging of tubes is specified, they shall remain in the respective sockets occupied during inspection of the equipment. Specified performance of the equipment shall be obtained without selection of tubes.

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3.6 Adjustment and repair.- The equipment shall be so constructed that parts, terminals, wiring, etc, are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number and variety of special tools, particularly those needed for tuning and adjustment. When tools for tuning or adjustment are needed, the unit where they are used shall provide convenient means for mounting them.

3.7 Interchangeability.- Corresponding components, replaceable subassemblies, and replaceable parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. (See 4.8) When dimensions, ratings, characteristics, etc, are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.8 Radio interference suppression.- The equipment shall conform to the requirements of Specification MIL-S-11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 3 feet from the equipment under test. (See 4.9)

3.9 Preconditioning.- Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4 (See also 4.3(d)).

3.10 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.2(a)(1)

3.11 Technical literature, tools and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.2(h) and (i))

3.12 Marking.-

3.12.1 General.- Marking shall conform to Specification MIL-M-13231, with the exception that the notation "US" shall be replaced with "Signal Corps, U. S. ARMY" on nameplate data for articles for equipment.

3.12.2 Visibility.- Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.12.3 Nameplates and serial numbers.- Modulator MD-22( )/URA-T1 and Mounting MT-250-( ) shall be provided with a nameplate and shall be serial numbered.

3.12.4 Identification marking of Cable Assemblies.- All Cable Assemblies listed in 1.1 shall be marked with identification bands in accordance with drawing SC-A-21825.



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3.12.5 Special Markings.-

3.12.5.1 Caution notice.- A caution plate, having a red background, shall be mounted on the front panel of the modulator immediately above the voltage indicator window: This notice shall read:

"WARNING

CHECK SETTING OF THE POWER SWITCHING PLUG BEFORE CONNECTING TO A POWER SOURCE"

3.12.6 Facsimilies.- Facsimilies as listed below shall be furnished conforming to Specification MIL-M-13231 and shall be submitted to the contracting officer for approval: (See 6.2j)

| | | Number of copies required |
|---|---|---|
| (a) | Nameplate data | 3 |
| (b) | Caution notice | 2 |
| (c) | Cable identification bands | 2 of each, 8 total |
| (d) | Voltage indicator plate | 2 |

3.13 Service conditions.- The equipment shall meet the following service conditions:

3.13.1 Equipment nonoperating.- Except as stated in 3.13.1(b), the equipment shall comply with the operational requirements of 3.13.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature.- Exposure in the range of + 160° F to -80° F; however, exposure at the high and low temperature extremes not to exceed 72 hours at any one time. (See 4.6.4)
(b) Relative humidity.- Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.13.2. (See 4.6.5)
(c) Elevation.- Elevation up to 50,000 feet above sea level.
(d) Orientation.- Indefinite storage in any orientation.
(e) Rain.- Two inches of rainfall per hour, with wind at nominal 20 miles per hour. (See 4.6.6)
(f) Vibration, bounce, and shock.- As described in 3.14.



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3.13.2 Equipment operating.- After a warmup period of 15 minutes, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Operation.- Continuous operation for a period of 1 year, alternating 6 hours on and 2 hours off, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction sample. (See 3.10)
(b) Temperature.- Ambient temperature in the range of 131°F to -40°F. (The 131°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours. (See 4.6.4)
(c) Relative humidity.- Relative humidity up to 95 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours. (See 4.6.5)
(d) Elevation.- Elevation up to 10,000 feet above sea level.
(e) Orientation.- Any orientation up to 20 degrees from normal operating orientation.

3.14 Vibration, bounce and shock.-

3.14.1 Vibration.- Except for internal resonance of specified parts and subassemblies, the Modulator MD-22( )/URA-T1 shall have no mechanical resonance below 55 cycles per second when shock mounts are blocked or removed. (See 4.6.1)

3.14.2 Bounce and shock.- The equipment shall meet the requirements of table I.

Table I. Bounce and shock

| Requirement | Inspection Paragraph | Performance after inspection |
|---|---|---|
| Bounce | 4.6.2 | Full specification performance. No physical damage except surface abrasion. |
| Shock; bench-handling | 4.6.3 | Full specification performance. No physical damage. |

3.15 Modulator MD-22( )/URA-T1.-



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3.15.1 Power input requirements.- The modulator shall be capable of being operated from either of the following primary power sources: (See 4.7.1.1.1 and 4.7.1.4)

(a) a-c input: 115 volts, 50 and 60 cycles
(b) d-c input:
  (1) 6 volts
  (2) 12 volts

3.15.2 Modes of operation.- The various modulator outputs shall be as specified below when the modulator is operated from a 115 volt, 60 cycle source. (See d-c operation 3.15.3)

3.15.2.1 Tones.- Audio frequency tones, as listed below, shall be provided in continuous succession, at the TRANSMITTER and OUTPUT jacks, by means of a motor diven commutator, when the SELECTOR switch is in the TONES position. The tones shall be within ±10% of the specified values when measured at 20°C(68°F). For each frequency the output level shall be as specified in 3.15.2.1.1. Clicks or noise in the output as a result of switching from one tone to another shall be held to a minimum.

| Commutator step | Frequency in cycles per sec |
|---|---|
| 1 | 525 |
| 2 | 700 |
| 3 | 650 |
| 4 | 575 |
| 5 | 750 |

In addition to the frequencies listed above, there shall be a tone of 450 cycles ± 10% present in the interval between each step. Also, a constant tone oscillator shall be provided which is adjustable to a frequency of 180 cycles per second. (See 4.7.1.1.2 and 4.7.1.1.3).

3.15.2.1.1 Output.-

(a) The signal output of each of the tones listed in 3.15.2.1 including the 450 and 180 cycle tones, shall be 0.20 volt rms ± 0.05 volt at the TRANSMITTER jack. Peak limiting shall be incorporated in the circuit to cut off the excessively high transient peaks. (See 4.7.1.1.4)
(b) The signal output of each of the tones shall be at least 0.6 watts at the OUTPUT jack, when measured with a test instrument which matches the impedance at the output circuit. (See 4.7.1.1.4)

3.15.2.2 Noise.- Random noise shall be generated and made available at the TRANSMITTER and OUTPUT jacks when the SELECTOR switch is in the NOISE position.



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3.15.2.2.1 Frequency response.- The frequency responseof the noise amplifier shall be flat within 3db between the minimum and maximum levels between the frequencies of 200 and 2500 cycles per second. The frequency response at 6000 cycles per second shall be at least 12db lower than the highest reading between 200 and 2500 cycles per second. (See 4.7.1.2.2)

3.15.2.2.2 Output.-

(a) The noise signal output at the TRANSMITTER jack shall be 0.3 volt ±0.05 volt.

(b) The noise signal output at the OUTPUT jack shall be at least 0.6 watts when measured with a test instrument which matches the impedance of the output circuit. (See 4.7.1.2.3)

3.15.2.3 MCW and CW.-

3.15.2.3.1 Keying.- Keying, in the MCW and CW modes of operation, shall be accomplished by the use of relaxation type keyer oscillators producing random operation of the keying relay. A SPEED control shall provide adjustment of the keying rate over a range equivalent to 10 words per minute minimum and 20 words per minute maximum. Klicks or noise due to keying shall be held to a minimum. (See 4.7.1.3.2)

3.15.2.3.2 MCW mode.- When the SELECTOR switch is in the MCW position, the output from the OUTPUT and TRANSMITTER jacks shall be a randomly interrupted audio signal. The pitch of the audio signal shall be variable over the range of operation of the pitch control. (See 4.7.1.3.3)

3.15.2.3.3 CW mode.- When the SELECTOR switch is in the CW position, the "tip-sleeve" circuit of the TRANSMITTER jack shall alternately make and brake to produce random CW keying of a radio transmitter.

3.15.3 d-c operation.- When the modulator is operated from 6- or 12 volt d-c supply the various outputs shall be as specified in 3.15.2 through 3.15.2.3.3 with the exception that there shall be no signal available at the OUTPUT jack. (See 4.7.1.4)

3.15.4 Commutator.-

3.15.4.1 Dependability.- The commutator shall be capable of continuous operation for a period of not less than 10 days (240 hours) with no appreciable wear or failure. (See 4.7.1.5)

3.15.4.2 Operation.- The commutator shall operate freely at temperatures between +131°F and -40°F.

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3.15.5 Power supply.- A power supply shall be included as an integral part of the modulator. It shall provide the necessary power for operation of the modulator to meet the requirements of this specification with power input variations as follows: 105 to 125 volts, 50 to 70 cycles; 5.0 to 7.0 volts dc; and 10.5 to 14.0 volts dc.

3.15.6 Housing.- The modulator shall be housed in a watertight enclosure capable of meeting the rain test requirements of 4.6.6.

3.15.6.1 Mounting facilities.- The housing shall fit on Mounting FT-250-( ).

3.15.6.2 Compartment.- A compartment of suitable size shall be provided at the rear of the housing for carrying accessories, such as one of each cord listed in 3.16.

3.15.7 Finish protective.- The modulator and mounting shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware type items such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on type I surfaces shall be gray color, semigloss enamel matching a color chip provided by the procuring agency. (See 6.4)

3.16 Cable Assemblies.- Assembly of all cables shall meet the requirements of Specification MIL-C-3885. (See 4.7.2)

3.16.1 Electrical Special Purpose Cable Assembly CX-2950/U.- Cable Assembly CX-2950/U, in accordance with Drawing SC-DL-184371, shall be a two-conductor, unshielded, ten-foot length of cordage type CO8-2(18) in accordance with Specification MIL-C-3884, with Plug PJ-055B on one end and PJ-068 on the other end. This cable will be used to connect from the TRANSMITTER jack of the Modulator to the key jack of the radio transmitter.

3.16.2 Electrical Special Purpose Cable Assembly CX-2951/U.- Cable Assembly CX-2951/U, in accordance with Drawing SC-DL-184372, shall be a two conductor, shielded, ten-foot length of cordage type WS-2 in accordance with Specification MIL-C-3883, with a Plug PJ-068 on each end. The cord will be used to connect from the TRANSMITTER jack of the modulator to the microphone jack of a radio transmitter.

3.16.3 Electrical Power Cable Assembly CX-3099/U.- Cable Assembly CX-3099/U, in accordance with Drawing SC-DL-184373, shall be a two conductor, ten-foot length of cordage type CO-02HOF(2/14)0535 in accordance with MIL-C-3432 with connector No. 92-F1 as made by American Phenolic Corporation, Chicago, Illinois, or equal, on one end and two battery type clips No. 24-A as made by Miller Electric Company, Cleveland, Ohio, or equal, on the other end. This cable will be used to couple the modulator to a 6-volt or 12-volt battery.



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3.16.4 Electrical Power Cable Assembly CX-111/URA-T1.- Cable Assembly CX-111/URA-T1, in accordance with Drawing SC-DL-184374, shall be a two conductor, six-foot length of cordage type CO-02LGF(2/18)0250 in accordance with Specification MIL-C-3432 with a connector type UP-120-M in accordance with Specification MIL-C-3767 on one end and a connector No.92-F1 as made by American Phenolic Corporation, or equal, on the other end. This cord shall be used to connect the modulator to an AC supply line.

3.17 Mounting FT-250-( ).- The Mounting FT-250-( ) shall be manufactured in accordance with Drawing and Data list SC-DL-14643 with the exception that the final paint film shall be as specified in 3.15.7. (See4.7.3)

3.18 Cover BG-185-( ).- The Cover BG-185-( ) shall be constructed in accordance with Drawing SC-D-1261 and shall fit over the modulator mounted on Mounting FT-250-( ) before and after application of shrinkage test of 4.7.4.1.

3.19 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portion of the following paragraphs in Specification MIL-P-11268:

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing

Connections, general
Grounding, general
Shielding on wire and cable
Flux and cleaning agents for soldering
Processes for soldering
Cleaning of equipment
General requirements for securing of parts
Mechanical operation of controls
Cleaning of welding
Processes for welding

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2)
(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.10.)

(1) Procurement inspection of items before preparation for delivery. (See 4.3.)
(2) Procurement inspection of preparation for delivery. (See 4.10 and section 5.)



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4.2 Preproduction inspection.- This inspection shall consist of the preproduction inspection specified in table II and the group A, group B, and group C inspection specified in tables III, IV, and V, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock, benchhandling; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II.- Preproduction inspection

| Inspection (For additional preproduction inspection see 4.2) | Rqt. par. | Insp par. |
|---|---|---|
| Vibration | 3.14.1 | 4.6.1 |
| Bounce | 3.14.2 | 4.6.2 |
| Rain | 3.13.1 | 4.6.6 |
| Moisture resistance | 3.13.1 and 3.13.2 | 4.6.5 |

4.3 Procurement inspection of items before preparation for delivery.- The minimum inspection performed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.8) Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A, group B, and group C inspection as specified in 4.3.1 through 4.3.3.2.

(b) When Standard MIL-STD-105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.9)



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4.3.1 Group A inspection.- This inspection ( including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be performed at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table III Group A inspection

| Inspection | Rqt. Par | Insp. Par. | AQL Major | AQL Minor |
|---|---|---|---|---|
| | | 4.5 | | |
| Visual and Mechanical | | | 4.0dphu | 15.0dphu |
| Modulator MD-22( )/URA-T1 | 3.15 | | 1.0% | 4.0% |
| Cable Ass'y CX-2950/U | 3.16.1 | | 1.0% | 4.0% |
| Cable Ass'y CX-2951/U | 3.16.2 | | 1.0% | 4.0% |
| Cable Ass'y CX-3099/U | 3.16.3 | | 1.0% | 4.0% |
| Cable Ass'y CX-111/URA-T1 | 3.16.4 | | 1.0% | 4.0% |
| Housing | 3.15.6 through 3.15.6.2 | | 1.0% | 4.0% |
| Mounting FT-250-( ) | 3.17 | | 1.0% | 4.0% |
| Cover BC-185-( ) | 3.18 | | 1.0% | 4.0% |
| Electrical | | | | |
| Modulator, MD-22( )/URA-T1 | 3.15 | 4.7.1 | | |
| Tones | 3.15.2.1 | 4.7.1.1 | | |
| Frequencies | 3.15.2.1 | 4.7.1.1.2 & 4.7.1.1.3 | 2.5 dphu for group combined | * |
| | 3.15.2.11 | 4.7.1.1.4 | | |
| Output | 3.15.2.2 | 4.7.1.2 | | |
| Noise | 3.15.2.2.1 | 4.7.1.2.2 | | |
| Frequency response | 3.15.2.2.2 | 4 7.1.2.3 | | |
| Output | 3.15.2.3 | 4.7.1.3 | | |
| MCW and CW | 3.15.2.3.1 | 4.7.1.3.2 | | |
| Keying speed | 3.15.2.3.2 | 4.7.1.3.3 | | |
| MCW | 3.16 | 4.7.2 | 1.0% | |
| Cable assemblies | | | | |

* All electrical defects are considered major.

4.3.2 Group B inspection.- This inspection (including sampling) shall conform to table IV and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL-STD-105 for disposition of rejected product.



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4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table IV, shall be as follows:

| Group B plan | AQL | Inspection level for normal inspection | Inspection level for reduced inspection |
|---|---|---|---|
| B-2 | 6.5% | L-7 | L-5 |

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

4.3.2.3 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table IV Group B inspection

| Inspection | Rqt Par | Insp Par | Group B plans (See 4.3.2.1) |
|---|---|---|---|
| Shock bench-handling | 3.14.2 | 4.6.3 | B-2 |
| Interchangeability | 3.7 | 4.8 | B-2 |
| d-c Operation (6-and 12-V.d.c.) | 3.15.3 | 4.7.1.4 | B-2 |

4.3.3 Group C inspection.- This inspection shall be as listed in table V.

4.3.3.1 Sampling for Group C inspection.-

(1) Commutator dependability.- One unit from first production lot and thereafter one unit of every 100 produced, or fraction thereof.

(2) Interference reduction.- One unit from first production lot and thereafter one unit out of every 100 produced, or fraction thereof.

(3) Shrinkage of Cover BG-185-( ).- One cover from first production lot and thereafter one cover out of every 200 bags produced, or fraction thereof.

(4) Temperature cycling.- One unit from first production lot and thereafter one unit out of every 50 produced, or fraction thereof.



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4.3.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5)

Table V.- Group C inspection

| Inspection | Rqt par. | Insp par. |
|---|---|---|
| Commutator dependability | 3.15.4.1 | 4.7.1.5 |
| Interference reduction | 3.8 | 4.9 |
| Shrinkage of Cover BC-185-() | 3.18 | 4.7.4.1 |
| Temperature cycling | 3.13.1 & 3.13.2 | 4.6.4 |

4.3.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.5 Disposition of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A, or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Bounce preconditioning.- The unit, with shock mounts blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.



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4.5 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VI.

Table VI-Classification of visual and mechanical defects

| Classification | Defect |
|---|---|
| Major | Gaskets on front panel and rear compartment deformed or otherwise defective in such a manner that they do not form a water tight seal.<br><br>Jack cover gaskets missing.<br><br>Front panel, bottom, and rear compartment door catches improperly located or defective. Front panel and rear compartment door catches do not pull the respective parts together tight enough to effect a water-tight seal.<br><br>Modulator case does not fit properly on Mounting FT-250-( ).<br><br>Controls inoperative.<br><br>Mounting holes or fastening hooks of Mounting FT-250-( )do not mate properly with mating parts of modulator.<br><br>Cover BG-185( ) does not fit properly over modulator and mounting. |
| Minor | Gaskets loose or not properly secured.<br><br>Catches extremely difficult to operate by hand.<br><br>Controls do not operate smoothly and freely. |

4.6 Service condition tests.-

4.6.1 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.14.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration.



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4.6.1 Contd

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel re-respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

4.6.2 Bounce test.- The Modulator, MD-22( )/URA-T1, on Mounting FT-250-( ), shall be secured to the Vehicular Adapter Plate and placed on the table of the Package Tester, both as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The Package Tester, shafts in phase, shall be operated at a speed of 285 rpm ±1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90 degrees, in the same direction, at the end of each 3/4-hour period.

4.6.3 Shock test; bench-handling.- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, the permit the assembly to drop back freely to the horizontal. Repeat, using other p practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.



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4.6.4 Temperature test for equipment.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Measurements shall be taken with power inputs of 115 volts a-c, 6-volts d c and 12-volts d-c at steps 1,4,5,8 and 10 as indicated: in table VII.

Table VII-Temperature cycling tests

| Step | Paragraph | Measurement | Test results |
|---|---|---|---|
| 1 | 4.7.1 through 4.7.1.4 | All operational tests | Requirements of 3.15 through 3.15.3 and 3.15.4.2* |
| 4 | 4.7.1.1 through 4.7.1.1.3, and 4.7.1.3.3 | Tone, and MCW | Requirements of 3.15.2.1, 3.15.2.3 through 3.15.2.3.2, and 3.15.4.2* |
| 5 | same as step 4 ( See 4.6.4.1) | same as step 4 | same as step 4 |
| 8 | same as step 4 | same as step 4 | same as step 4 |
| 10 | same as step 1 | same as step 1 | same as step 1 |

*NOTE: The commutator operation shall be checked throughout the cycle to determine compliance with 3.15.4.2.

4.6.4.1 Altitude test.- This test shall be conducted at step 5 of the temperature cycling test of 4.6.4 to check compliance with 3.13.2(d). Immediately after performing the measurements at step 5, and with the equipment still operating, the pressure in the temperature chamber shall be reduced to a pressure of 20.6 inches of mercury (approximately equal to 10,000 ft elevation). The measurements of 4.7.1.1 through 4.7.1.1.3, and 4.7.1.3.3. shall again be made. Operation shall be as indicated in step 5 of Table VII.



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4.6.5 Moisture-resistance test for equipment.- The equipment shall be tested as follows:

4.6.5.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.
(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.
(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.6.5.2 Test method.-

(a) Dry at 130° ±5° F for 24 hours.
(b) Condition at 77° ±5° F and 40 to 50 percent relative humidity for 24 hours.
(c) Take measurements as specified in 4.6.5.3, and readjust or realine as necessary to meet full specification requirements.
(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.6.5.3 shall be made during the periods shown on the standard and shall comply with 4.6.5.3.
(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.6.5.3.

4.6.5.3 Performance.- Measurements made during test, and specified performance during cycling, shall be as follows:

| Paragraph | Measurement | Performance during cycling |
|---|---|---|
| 4.7.1.1 through 4.7.1.1.3 | Tones | Requirements of 3.15.2.1 |
| 4.7.1.3.3 | MCW | Requirements of 3.15.2.3 through 3.15.2.3.2 |

4.6.6 Rain test.- The component shall be tested, step by step, as follows:

4.6.6.1 Dry at 150° F for 48 hours.

4.6.6.2 Condition for 4 hours at 77° ±5° F and 40 to 60 percent relative humidity.

4.6.6.3 Take the following measurements to insure that the component meets specified performance:

| Paragraph | Measurement |
|---|---|
| 4.7.1.1 through 4.7.1.1.3 | Tones |
| 4.7.1.3.3 | MCW |



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4.6.6.4 Place the component, in its normal operating position, in a test chamber maintained at 77° ±5° F. Allow simulated rainfall to fall on exposed surfaces of the component. The water shall be at a temperature of 65° ±5° F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.6.6.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the component. The water shall fall on exposed surfaces and be blown by the wind against one side of the component. Maintain this condition for 1 hour.

4.6.6.6 Rotate the component 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

4.6.6.7 Continue as in 4.6.6.6, until all four sides of the component have been exposed to the wind and rain, each side for 1 hour.

4.6.6.8 Immediately after completion of the foregoing, repeat the measurement specified in 4.6.6.3, without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The measurements shall meet specified performance.

4.6.6.9 Open the component. There shall be no visible evidence of leakage.

4.7 Performance tests.-

4.7.1 Modulator MD-22( )/URA-T1.-

4.7.1.1 Tones.- The modulator shall be operated in the Tone mode and shall be tested as indicated below, for compliance with 3.15.2.1 and 3.15.2.1.1.

4.7.1.1.1 Conditions.- The tests shall be made with power input of 115 volts, 60 cycles, at an ambient temperature of 20°C(68°F) unless otherwise stated.

4.7.1.1.2 Commutator steps.- With the selector switch turned to the TONES position and the background oscillator tube removed, the output signal frequency at each commutator step, and the signal frequency between steps, shall be measured at the TRANSMITTER jack using Frequency Meter FR-67/U or equal. The frequency of the tones shall meet the requirements of 3.15.2.1: (Note: The commutator motor may be disconnected and the commutator turned by hand to facilitate measurement of the individual tones.)

4.7.1.1.3 Background tone.- With the background oscillator tube replaced and the AF oscillator disabled, the background tone frequency shall be measured. The background tone control shall be adjusted for a 180 cycle tone as specified in 3.15.2.1: The control shall be near the center of its range.

4.7.1.1.4 Output.-

(a) The signal voltage shall be measured at the TRANSMITTER jack using Voltmeter ME-30A/U or equal shunted with 100 ohms. This voltage shall be as specified in 3.15.2.1.1 (a):



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(b) The audio power output shall be measured at the OUTPUT using a suitable audio power meter which matches the impedance of the output. The audio power of each tone shall be as specified in 3.15.2.1.1(b).

4.7.1.2 Noise.- The modulator shall be operated in the noise mode and shall be tested as indicated below to determine compliance with 3.15.2.2 through 3.15.2.2.2.

4.7.1.2.1 Conditions.- The conditions for these tests shall be as indicated in 4.7.1.1.1.

4.7.1.2.2 Noise amplifier frequency response.- The frequency response of the noise amplifier shall be measured over the frequency range of 200 to 6000 cycles with the noise generator tube and the noise control tube removed. An Audio Oscillator, TS-382/U or equal, shall be fed into the plate connection of the noise generator tube socket with the output maintained at 0.04 volt throughout the range of measurement. Measurements for a response curved shall be made at the TRANSMITTER jack, and shall be as specified in 3.15.2.2.1.

4.7.1.2.3 Noise Output.- The noise signal output shall be measured with the noise generator and noise control tubes replaced. The output at the TRANSMITTER and OUTPUT jacks shall be as specified in 3.15.2.2.2(a) and 3.15.2.2.2(b) when measured using the procedure described in 4.7.1.1.4.

4.7.1.3 MCW and CW.- The modulator shall be operated in the MCW and CW modes and tested as indicated below to determine compliance with 3.15.2.3 through 3.15.2.3.3.

4.7.1.3.1 Conditions.- The conditions for these test shall be as indicated in 4.7.1.1.1.

4.7.1.3.2 Keying speed.- The CW keying output from the TRANSMITTER jack shall be monitored using a suitable code recorder, and the keying speed shall be calculated at the maximum and minimum settings of the SPEED Control to determine compliance with 3.15.2.3.1. The number of words per minute shall be based on the following:

(a) Three dots or dashes equal one letter.
(b) Five letters equal one word.

4.7.1.3.3 MCW.- The operation of the PITCH control shall be checked by shorting pins 6 and 7 of the keying relay and monitoring the output with a headset connected to the OUTPUT jack. The PITCH control shall be varied throughout its range to determine compliance with 3.15.2.3.2. The relay short shall then be removed, after which, a randomly interrupted audio tone shall be observed.

4.7.1.4 D-C operation.- The tests of 4.7.1.1 through 4.7.1.3.3 shall be repeated with power inputs of 6-volts and 12-volts to check compliance with 3.15.3 and 3.15.5. The results of the tests shall be the same as in 4.7.1.1 through 4.7.1.3.3, with the exception that all references to the OUTPUT jack are not applicable.

NOTE: The MCW output check described in 4.7.1.3.3 can be made at the TRANSMITTER jack if a suitable adapter is used, allowing the headset to be connected to the 3 circuit jack.



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4.7.1.5 Commutator dependability check.- The commutator shall be operated continuously, from a 115 volt a-c source, for a period of 10 days, (240 hours) and then disassembled and inspected for compliance with 3.15.4.1. There shall be no evidence of appreciable wear or failure.

4.7.2 Cable assemblies.- Cable Assemblies CX-2950/U, CX-2951/U and CX-3099/U, and CX-111/URA-T1 shall be tested in accordance with the test procedures of Specification MIL-C-3885.

4.7.3 Mounting FT-250-( ).- Mounting FT-250-( ) shall be inspected to determine compliance with the requirements of SC-DL-14643 and 3.17, and shall be checked or gaged to determine fit on Modulator MD-22( )/URA-T1.

4.7.4 Cover BG-185-( ).- Cover BG-185-( ) shall be tested to determine compliance with the requirements of SC-D-12631 and 3.18.

4.7.4.1 Test for shrinkage.- The cover shall be inspected for satisfactory fit both before and after the following test: Soak the cover in tap water at normal room temperature for 24 hours and then dry thoroughly at normal room temperature.

4.8 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of the modulator housing and any insertable assemblies, when such dimensions affect mating of parts.
(b) Location of fasteners on the modulator housing which must mate, such as front panel, rear compartment door, and mounting fasteners.
(c) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.
(d) Size and form of special threads.
(e) Fit of Modulator MD-22( )/URA-T1 on Mounting FT-250-( ).
(f) Fit of Mounting FT-250-( ) to Modulator FT-250-( ).
(g) Fit of Cover BG-185-( ) on Modulator and Mounting.

4.9 Interference reduction.- Conduct interference reduction tests in accordance with the requirements and tests of Specification MIL-I-11748, Class I equipment, and record data for compliance with 3.8.



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4.10 Procurement inspection of preparation for delivery.- The minimum inspection performed by the contractor shall be as specified in 4.10.1 and 4.10.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.8) Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.10.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table VIII.

TABLE VIII

PRESERVATION, AND PACKING AND MARKING THEREOF

| MAJOR | MINOR |
|---|---|
| 1. Use of improper or defective material. | 1. Item not properly blocked or braced within the unit package to prevent movement. |
| 2. Quantity in unit package not as specified. | 2. Packaging material damaged. |
| 3. Incorrect packaging method applied. | 3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible. |
| 4. Cushioning or padding omitted. | |
| 5. Cushioning inadequate for the physical and mechanical protection of the item. | |
| 6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope. | |
| 7. Stock number omitted, incorrect, or illegible. | |
| 8. Nomenclature omitted, incorrect, or illegible. | |
| 9. Marking of quantity of items in package omitted, incorrect, or illegible. | |
| 10. Different stock numbered items in the same unit package. | |

4.10.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on Table IX.

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## TABLE IX

### PACKING, AND MARKING THEREOF

| MAJOR | MINOR |
|---|---|
| 1. Use of improper or defective material. | 1. Unsealed carton. |
| 2. Quantity in pack not as specified. | 2. Defective taping or sealing of carton. |
| 3. Gross weight in excess of specified amount. | 3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105. |
| 4. Box closure not as specified. | 4. Any item of required marking information other than 10 thru 16 listed under major defects omitted, incorrect, or illegible. |
| 5. Type, grade, class, and style of the shipping container not as specified. | |
| 6. Strapping omitted. | |
| 7. Strapping inadequate or incorrectly applied. | |
| 8. Items not adequately blocked, braced or cushioned within the shipping container to prevent movement or damage. | |
| 9. Shipping documents or packing list omitted. | |
| 10. Stock number omitted, incorrect, or illegible. | |
| 11. Nomenclature omitted, incorrect, or illegible. | |
| 12. Marking of quantity of items in pack omitted, incorrect, or illegible. | |
| 13. Destination marking omitted, incorrect, or illegible. | |
| 14. Service designation (color marking) omitted. | |
| 15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible. | |
| 16. Overseas code marking omitted, incorrect, or illegible. | |



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4.10.3 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.10.4 Precedure in case of failure.- If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.10.5 Disposition of nonconforming product.- Disposition of nonconforming [illegible] (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.10.6 Rough handling test (preparation for delivery).- When rough handling test is required by the contract (see 6.2 (c)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

| Paragraph | Test |
|---|---|
| 4.7.1.1.2 | Tones |
| 4.7.1.3.3 | MCW Tone |

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Training Generator AN/URA-71( ) (Components of) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Technical manual.- EAch technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting, type I, class b bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Tubes.- Remove all operating tubes over four inches in length not securely clamped in place and together with spare tubes, package individually, in accordance with the applicable provisions of Specification MIL-P-75.

5.1.1.3 Fuses.- Fuses shall be packaged Method III, as follows: Group fuses bearing the same stock number together and cushion by wrapping in flexible, single-face, corrugated fiberboard, type I, conforming to Specification LLL-F-291. Secure the cushioning with gummed paper tape, class 1, conforming to Specification UU-T-111.

5.1.1.4 Cover BG-185.- Each cover shall be packaged Method III as follows: Fold each cover into a bundle of proportionate dimensions and tie in two directions with cotton twine, 6 ply, type II, conforming to Specification T-T-871.



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5.1.1.5 Electrical Special purpose cable assemblies CX-2950/U and CX-2951/U and Electrical Power Cable A sembly CX-3099/U.- Each cable assembly shall be packaged method III as follows: Wind each cable into a coil of proportionate dimensions and tie in three places with cotton twine, securing the connectors to the coils with the ties. Cushion each cable by wrapping in type II, single face, flexible corrugated fiberboard, conforming to Specification LLL-F-291. Secure the cushioning with class 2, gummed paper tape, conforming to Specification UU-T-111.

5.1.1.6 Cord CX-111/URA-T1.- Each cord shall be packaged Method III as follows: Wind each cord into a coil of proportionate dimensions and tie in three places with cotton twine, securing the connectors to the coil with the ties.

5.1.1.7 Consolidated package.- Consolidate the items packaged as specified in 5.1.1.2 through 5.1.1.5 together within a close-fitting, regular slotted style, class 1, W5c box, conforming to Specification PPP-B-636. Fill all voids with type I, single face, flexible corrugated fiberboard. Box closure shall be in accordance with the requirements of the box specification.

5.1.1.8 Modulator MD-22( )/URA-T1 on Mounting FT-250( ).-

5.1.1.8.1 Stowage.- Stow the cord CX-111/URA-T1, packaged as specified in 5.1.1.6 within the designated compartment of the modulator case. Close case and secure fastenings.

5.1.1.8.2 Packaging of modulator on mounting.- Each modulator on mounting shall be packaged IC-5 as follows: Cushion each modulator together with mounting on all surfaces with cells pads or both, fabricated of corrugated fiberboard, having a minimum dry bursting strength of 275 pounds, designed to absorb the shock of impact encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned modulator with mounting, together with consolidated package, packaged as specified in 5.1.1.7 within a close-fitting, regular soltted style, class 2, W5e box, conforming to Specification PPP-B-636. Box closure shall be effected by sealing all seams and joints with water resistant pressure sensitive tape, conforming to Specification PPP-T-76.

5.1.2 Level C.- Training Generator AN/URA/T1( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Training Generator AN/URA-T1( ) shall be packed within a close-fitting style 4, nailed wood box, conforming to class 2 of Specification PPP-B-621. Place the packaged technical manual between the contents and the lid of the shipping container. Closure shall be in accordance with the box specification.

5.2.1.1 Metal strapping.- Shipping container shall be strapped in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.



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5.2.2 Level B.- Training Generator AN/URA-T1( ) shall be packed as specified in 5.2.1, except that the style 4, nailed wood box shall conform to class 1 of Specification PPP-B-621.

5.2.3 Level C.- Training Generator AN/URA-T1( ) shall be packed for shipment in a manner, conforming to the requirements of Uniform Freight Classification No.4 and Consolidated Freight Classification Rule No.21 for rail shipment, National Motor Freight Classification No.A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Training Generator AN/URA-T1( ) is intended for use in training radio communications operators to read code through various types of jamming. This generator may be used to key or modulate Signal Corps communications transmitters producing intentional interference of the following characteristics:

(a) Tones.- A repeated series of five notes creating a sound similar to the sound of bagpipes with a 450-cycle tone repeated in the interval between the five notes, and with a 180-cycle tone continuously superimposed.
(b) Noise.- A hissing, crackling sound similar static noise heard in a receiver operating near high tension lines or sparking electrical equipment.
(c) Random keying.- A continuous series of meaningless dots and dashes which will either key an audio tone which is fed into a transmitter microphone circuit, or key the CW circuit of a communications transmitter.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C.)
(d) The specific paragraphs section 5 which are applicable to the particular procurement.
(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.10.)



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(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.
d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.
e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required. (See 3.11)
(i) Quantity of tools and running spare parts required. (See 3.11)
(j) Facsimiles required. (See 3.12.6)
(k) Submission of the statement of treatment referenced in 3.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.
(l) Source of color chips. (See 3.15.7)

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PIQ-1W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Color.- The color chip furnished by the procuring agency will match color chip No. 2610 of superseded Specification TT-C-595. No matching color chip has been provided in the superseding Federal Standard No. 595. (See 3.15.7)

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion fo the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.2)

6.6 Definition of Level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.



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6.7.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.7.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.8 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL-STD-105.

(b) Type B---That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL-STD-105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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